

Tain/Val Split on all sites except NYU (test)

--lr 0.0001 --wd 0.0001 --batch 8 --epochs 15 (no layers frozen)

Epoch 1 [Train]: 100%|████████████████████| 82/82 [00:20<00:00, 3.97it/s]

Epoch 1 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 3.86it/s]

Epoch 1 Summary:

Train Loss: 1.1015 | Acc: 0.5076 | F1: 0.5099 | AUC: 0.5192

Val Loss: 0.8993 | Acc: 0.5342 | F1: 0.3818 | AUC: 0.5110

>>> Saved new best model at epoch 1 with AUC 0.5110

Epoch 2 [Train]: 100%|████████████████████| 82/82 [00:17<00:00, 4.67it/s]

Epoch 2 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.39it/s]

Epoch 2 Summary:

Train Loss: 0.8138 | Acc: 0.5564 | F1: 0.5474 | AUC: 0.5882

Val Loss: 0.8030 | Acc: 0.5411 | F1: 0.5379 | AUC: 0.5234

>>> Saved new best model at epoch 2 with AUC 0.5234

Epoch 3 [Train]: 100%|████████████████████| 82/82 [00:17<00:00, 4.71it/s]

Epoch 3 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.42it/s]

Epoch 3 Summary:

Train Loss: 0.7270 | Acc: 0.6082 | F1: 0.6003 | AUC: 0.6398

Val Loss: 0.8802 | Acc: 0.5479 | F1: 0.3654 | AUC: 0.5360

>>> Saved new best model at epoch 3 with AUC 0.5360

Epoch 4 [Train]: 100%|████████████████████| 82/82 [00:17<00:00, 4.60it/s]

Epoch 4 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.43it/s]

Epoch 4 Summary:

Train Loss: 0.7171 | Acc: 0.6006 | F1: 0.5854 | AUC: 0.6365

Val Loss: 0.7485 | Acc: 0.5822 | F1: 0.5197 | AUC: 0.5541

>>> Saved new best model at epoch 4 with AUC 0.5541

Epoch 5 [Train]: 100%|████████████████████| 82/82 [00:17<00:00, 4.61it/s]

Epoch 5 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.61it/s]

Epoch 11 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.33it/s]

Epoch 11 Summary:

Train Loss: 0.5363 | Acc: 0.7302 | F1: 0.7186 | AUC: 0.8042

Val Loss: 0.8949 | Acc: 0.4795 | F1: 0.6042 | AUC: 0.6029

>>> Saved new best model at epoch 11 with AUC 0.6029

Epoch 12 [Train]: 100%|████████████████████| 82/82 [00:17<00:00, 4.64it/s]

Epoch 12 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.55it/s]

Epoch 12 Summary:

Train Loss: 0.4951 | Acc: 0.7546 | F1: 0.7542 | AUC: 0.8407

Val Loss: 0.7461 | Acc: 0.5890 | F1: 0.4643 | AUC: 0.5865

Epoch 13 [Train]: 100%|████████████████████| 82/82 [00:17<00:00, 4.67it/s]

Epoch 13 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.52it/s]

Epoch 13 Summary:

Train Loss: 0.4559 | Acc: 0.8110 | F1: 0.8080 | AUC: 0.8706

Val Loss: 0.9093 | Acc: 0.5274 | F1: 0.1882 | AUC: 0.5641

Epoch 14 [Train]: 100%|████████████████████| 82/82 [00:18<00:00, 4.53it/s]

Epoch 14 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.50it/s]

Epoch 14 Summary:

Train Loss: 0.4254 | Acc: 0.7973 | F1: 0.7925 | AUC: 0.8887

Val Loss: 0.7792 | Acc: 0.5959 | F1: 0.6144 | AUC: 0.6151

>>> Saved new best model at epoch 14 with AUC 0.6151

Epoch 15 [Train]: 100%|████████████████████| 82/82 [00:17<00:00, 4.60it/s]

Epoch 15 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.65it/s]

Epoch 15 Summary:

Train Loss: 0.3764 | Acc: 0.8430 | F1: 0.8398 | AUC: 0.9202

Val Loss: 0.7940 | Acc: 0.6370 | F1: 0.5138 | AUC: 0.6373

>>> Saved new best model at epoch 15 with AUC 0.6373

Training log saved to: ./ABIDEI/training_log.csv

ROC curve saved to: ./ABIDEI/val_roc_curve.png

TEST SET:

F1-score: 0.35384615384615387

ROC_AUC-score: 0.5242210321324245

Accuracy: 54.10%

```
Trainable parameters:
classifier.conv_6.weight
classifier.conv_6.bias
```

```
Epoch 1 [Val]: 100%|███████████| 19/19 [00:05<00:00,  
3.71it/s]
```

```
Epoch 2 [Val]: 100%|██████████████████████| 19/19 [00:04<00:00,  
4.37it/s]
```

```
Epoch 3 [Val]: 100%|███████████| 19/19 [00:04<00:00,  
4.29it/s]
```

```
Epoch 4 [Val]: 100%|███████████| 19/19 [00:04<00:00,  
4.44it/s]
```

```
>>> Saved new best model at epoch 4 with AUC 0.5398
```

Epoch 5 [Train]: 100%|████████████████████| 82/82 [00:16<00:00, 5.06it/s]

Epoch 5 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.30it/s]

Epoch 5 Summary:

Train Loss: 1.4247 | Acc: 0.5198 | F1: 0.5532 | AUC: 0.5403

Val Loss: 1.4696 | Acc: 0.5000 | F1: 0.5466 | AUC: 0.5400

>>> Saved new best model at epoch 5 with AUC 0.5400

Epoch 6 [Train]: 100%|████████████████████| 82/82 [00:15<00:00, 5.40it/s]

Epoch 6 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.15it/s]

Epoch 6 Summary:

Train Loss: 1.3137 | Acc: 0.5320 | F1: 0.5694 | AUC: 0.5622

Val Loss: 1.4428 | Acc: 0.5068 | F1: 0.5443 | AUC: 0.5398

Epoch 7 [Train]: 100%|████████████████████| 82/82 [00:18<00:00, 4.49it/s]

Epoch 7 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.40it/s]

Epoch 7 Summary:

Train Loss: 1.3108 | Acc: 0.5412 | F1: 0.5837 | AUC: 0.5581

Val Loss: 1.4942 | Acc: 0.5068 | F1: 0.5556 | AUC: 0.5383

Epoch 8 [Train]: 100%|████████████████████| 82/82 [00:14<00:00, 5.60it/s]

Epoch 8 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.30it/s]

Epoch 8 Summary:

Train Loss: 1.3251 | Acc: 0.5213 | F1: 0.5639 | AUC: 0.5477

Val Loss: 1.4857 | Acc: 0.5068 | F1: 0.5556 | AUC: 0.5389

Epoch 9 [Train]: 100%|████████████████████| 82/82 [00:15<00:00, 5.47it/s]

Epoch 9 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.32it/s]

Epoch 9 Summary:

Train Loss: 1.3370 | Acc: 0.5381 | F1: 0.5774 | AUC: 0.5438

Val Loss: 1.4301 | Acc: 0.5000 | F1: 0.5350 | AUC: 0.5368

Epoch 10 [Train]: 100%|████████████████████| 82/82 [00:17<00:00, 4.59it/s]

Epoch 10 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.13it/s]

Epoch 10 Summary:

Train Loss: 1.3298 | Acc: 0.5488 | F1: 0.5854 | AUC: 0.5584

Val Loss: 1.5051 | Acc: 0.5137 | F1: 0.5697 | AUC: 0.5396

Epoch 11 [Train]: 100%|████████████████████| 82/82 [00:15<00:00, 5.16it/s]
Epoch 11 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.30it/s]

Epoch 11 Summary:

Train Loss: 1.3589 | Acc: 0.5396 | F1: 0.5817 | AUC: 0.5527

Val Loss: 1.4868 | Acc: 0.5137 | F1: 0.5644 | AUC: 0.5383

Epoch 12 [Train]: 100%|████████████████████| 82/82 [00:18<00:00, 4.46it/s]

Epoch 12 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.33it/s]

Epoch 12 Summary:

Train Loss: 1.3203 | Acc: 0.5381 | F1: 0.5798 | AUC: 0.5605

Val Loss: 1.4430 | Acc: 0.4932 | F1: 0.5375 | AUC: 0.5385

Epoch 13 [Train]: 100%|████████████████████| 82/82 [00:17<00:00, 4.77it/s]

Epoch 13 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.30it/s]

Epoch 13 Summary:

Train Loss: 1.3254 | Acc: 0.5549 | F1: 0.5817 | AUC: 0.5614

Val Loss: 1.4917 | Acc: 0.5068 | F1: 0.5556 | AUC: 0.5381

Epoch 14 [Train]: 100%|████████████████████| 82/82 [00:17<00:00, 4.71it/s]

Epoch 14 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.28it/s]

Epoch 14 Summary:

Train Loss: 1.3250 | Acc: 0.5290 | F1: 0.5678 | AUC: 0.5612

Val Loss: 1.4735 | Acc: 0.5068 | F1: 0.5500 | AUC: 0.5402

>>> Saved new best model at epoch 14 with AUC 0.5402

Epoch 15 [Train]: 100%|████████████████████| 82/82 [00:17<00:00, 4.68it/s]

Epoch 15 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.37it/s]

Epoch 15 Summary:

Train Loss: 1.2986 | Acc: 0.5549 | F1: 0.5887 | AUC: 0.5763

Val Loss: 1.4876 | Acc: 0.5000 | F1: 0.5466 | AUC: 0.5385

Training log saved to: ./ABIDEI/training_log.csv

ROC curve saved to: ./ABIDEI/val_roc_curve.png

Training complete.

--lr 0.0001 --wd 0.0001 --batch 8 --epochs 15 (last feature layer and classifier layer frozen)

Trainable parameters:

feature_extractor.conv_5.0.weight
feature_extractor.conv_5.0.bias
feature_extractor.conv_5.1.weight
feature_extractor.conv_5.1.bias
classifier.conv_6.weight
classifier.conv_6.bias

Epoch 1 [Train]: 100%|████████████████████| 82/82 [00:15<00:00, 5.24it/s]

Epoch 1 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.47it/s]

Epoch 1 Summary:

Train Loss: 1.3487 | Acc: 0.5274 | F1: 0.5571 | AUC: 0.5476

Val Loss: 1.4120 | Acc: 0.5000 | F1: 0.5466 | AUC: 0.5345

>>> Saved new best model at epoch 1 with AUC 0.5345

Epoch 2 [Train]: 100%|████████████████████| 82/82 [00:17<00:00, 4.66it/s]

Epoch 2 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.40it/s]

Epoch 2 Summary:

Train Loss: 1.2187 | Acc: 0.5290 | F1: 0.5666 | AUC: 0.5603

Val Loss: 1.3773 | Acc: 0.5068 | F1: 0.5556 | AUC: 0.5306

Epoch 3 [Train]: 100%|████████████████████| 82/82 [00:15<00:00, 5.36it/s]

Epoch 3 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.37it/s]

Epoch 3 Summary:

Train Loss: 1.2035 | Acc: 0.5488 | F1: 0.5819 | AUC: 0.5687

Val Loss: 1.3125 | Acc: 0.5000 | F1: 0.5350 | AUC: 0.5283

Epoch 4 [Train]: 100%|████████████████████| 82/82 [00:16<00:00, 5.00it/s]

Epoch 4 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 4.46it/s]

Epoch 4 Summary:

Train Loss: 1.2274 | Acc: 0.5305 | F1: 0.5625 | AUC: 0.5458

Val Loss: 1.2952 | Acc: 0.5068 | F1: 0.5443 | AUC: 0.5236

```
Epoch 5 [Train]: 100%|██████████| 82/82 [00:15<00:00, 5.39it/s]
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```
Epoch 5 [Val]: 100%|███████████| 19/19 [00:04<00:00,  
4.39it/s]
```

Epoch 5 Summary:

Train Loss: 1.2450 | Acc: 0.5274 | F1: 0.5468 | AUC: 0.5402

Val	Loss: 1.2428	Acc: 0.5000	F1: 0.5290	AUC: 0.5140
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```
Epoch 6 [Train]: 100%|██████████| 82/82 [00:16<00:00,  
4.98it/s]
```

```
Epoch 6 [Val]: 100%|███████████| 19/19 [00:05<00:00,  
3.72it/s]
```

Epoch 6 Summary:

Train Loss: 1.1372 | Acc: 0.5290 | F1: 0.5476 | AUC: 0.5546

Val	Loss: 1.1975	Acc: 0.4932	F1: 0.5132	AUC: 0.5121
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Epoch 7 [Train]: 100%|██████████| 82/82 [00:19<00:00,  
4.31it/s]
```

```
Epoch 7 [Val]: 100%|██████████████████████| 19/19 [00:04<00:00,  
4.19it/s]
```

Epoch 7 Summary:

Train Loss: 1.1204 | Acc: 0.5366 | F1: 0.5543 | AUC: 0.5506

Val Loss: 1.1850 | Acc: 0.4795 | F1: 0.5000 | AUC: 0.5086

```
Epoch 8 [Train]: 100%|██████████| 82/82 [00:15<00:00,  
5.24it/s]
```

```
Epoch 8 [Val]: 100%|██████████████████████| 19/19 [00:04<00:00,  
4.39it/s]
```

Epoch 8 Summary:

Train Loss: 1.0955 | Acc: 0.5229 | F1: 0.5335 | AUC: 0.5489

Val	Loss: 1.1725	Acc: 0.4863	F1: 0.5098	AUC: 0.5057
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```
Epoch 9 [Train]: 100%|██████████| 82/82 [00:16<00:00,  
4.93it/s]
```

```
Epoch 9 [Val]: 100%|██████████████████████| 19/19 [00:04<00:00,  
4.38it/s]
```

Epoch 9 Summary:

Train Loss: 1.1212 | Acc: 0.5335 | F1: 0.5378 | AUC: 0.5364

Val	Loss: 1.1474	Acc: 0.4658	F1: 0.4583	AUC: 0.5039
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```
Epoch 10 [Train]: 100%|██████████| 82/82 [00:15<00:00,  
5.28it/s]
```

```
Epoch 10 [Val]: 100%|██████████| 19/19 [00:04<00:00,  
4.48it/s]
```

Epoch 10 Summary:

Train Loss: 1.0761 | Acc: 0.5412 | F1: 0.5446 | AUC: 0.5604
Val Loss: 1.1377 | Acc: 0.4795 | F1: 0.4933 | AUC: 0.5029
Epoch 11 [Train]: 100%|████████████████████| 82/82 [00:17<00:00,
4.82it/s]
Epoch 11 [Val]: 100%|████████████████████| 19/19 [00:05<00:00,
3.67it/s]

Epoch 11 Summary:

Train Loss: 1.1134 | Acc: 0.5137 | F1: 0.5130 | AUC: 0.5360
Val Loss: 1.1007 | Acc: 0.4521 | F1: 0.4366 | AUC: 0.4995
Epoch 12 [Train]: 100%|████████████████████| 82/82 [00:19<00:00,
4.12it/s]
Epoch 12 [Val]: 100%|████████████████████| 19/19 [00:04<00:00,
4.26it/s]

Epoch 12 Summary:

Train Loss: 1.0606 | Acc: 0.5366 | F1: 0.5408 | AUC: 0.5584
Val Loss: 1.0930 | Acc: 0.4589 | F1: 0.4397 | AUC: 0.4975
Epoch 13 [Train]: 100%|████████████████████| 82/82 [00:15<00:00,
5.44it/s]
Epoch 13 [Val]: 100%|████████████████████| 19/19 [00:04<00:00,
4.35it/s]

Epoch 13 Summary:

Train Loss: 1.0595 | Acc: 0.5396 | F1: 0.5296 | AUC: 0.5562
Val Loss: 1.0927 | Acc: 0.4658 | F1: 0.4429 | AUC: 0.4961
Epoch 14 [Train]: 100%|████████████████████| 82/82 [00:15<00:00,
5.35it/s]
Epoch 14 [Val]: 100%|████████████████████| 19/19 [00:04<00:00,
4.32it/s]

Epoch 14 Summary:

Train Loss: 1.0503 | Acc: 0.5534 | F1: 0.5581 | AUC: 0.5585
Val Loss: 1.0856 | Acc: 0.4589 | F1: 0.4397 | AUC: 0.4992
Epoch 15 [Train]: 100%|████████████████████| 82/82 [00:15<00:00,
5.39it/s]
Epoch 15 [Val]: 100%|████████████████████| 19/19 [00:04<00:00,
4.49it/s]

Epoch 15 Summary:

Train Loss: 1.0268 | Acc: 0.5473 | F1: 0.5507 | AUC: 0.5723
Val Loss: 1.0769 | Acc: 0.4658 | F1: 0.4429 | AUC: 0.4971
Training log saved to: ./ABIDEI/training_log.csv
ROC curve saved to: ./ABIDEI/val_roc_curve.png

Training complete.

```
--lr 0.0001 --wd 0.0001 --batch 8 --epochs 15 (no layers frozen)
```

```
Epoch 1 [Val]: 100%|██████████████████████| 4/4 [00:01<00:00,  
2.27it/s]
```

```
Epoch 2 [Val]: 100%|██████████| 4/4 [00:01<00:00,  
2.27it/s]
```

```
Epoch 3 [Val]: 100%|██████████████████████| 4/4 [00:01<00:00,  
2.30it/s]
```

```
Epoch 4 [Val]: 100%|██████████████████████| 4/4 [00:01<00:00,  
2.28it/s]
```

```
Epoch 5 [Val]: 100%|██████████████████████| 4/4 [00:01<00:00,  
2.23it/s]
```

Val	Loss: 0.6560	Acc: 0.5926	F1: 0.4211	AUC: 0.7000
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Epoch 6 [Train]: 100%|████████████████████| 16/16 [00:06<00:00, 2.45it/s]
Epoch 6 [Val]: 100%|████████████████████| 4/4 [00:01<00:00, 2.24it/s]

Epoch 6 Summary:

Train Loss: 0.5145 | Acc: 0.7266 | F1: 0.6465 | AUC: 0.8204

Val Loss: 0.6506 | Acc: 0.7407 | F1: 0.7200 | AUC: 0.7111

>>> Saved new best model at epoch 6 with ACC 0.7407

Epoch 7 [Train]: 100%|████████████████████| 16/16 [00:06<00:00, 2.47it/s]

Epoch 7 [Val]: 100%|████████████████████| 4/4 [00:01<00:00, 2.29it/s]

Epoch 7 Summary:

Train Loss: 0.4152 | Acc: 0.7969 | F1: 0.7636 | AUC: 0.8879

Val Loss: 0.7349 | Acc: 0.5926 | F1: 0.3529 | AUC: 0.7278

Epoch 8 [Train]: 100%|████████████████████| 16/16 [00:06<00:00, 2.43it/s]

Epoch 8 [Val]: 100%|████████████████████| 4/4 [00:01<00:00, 2.24it/s]

Epoch 8 Summary:

Train Loss: 0.3296 | Acc: 0.8828 | F1: 0.8624 | AUC: 0.9432

Val Loss: 0.6909 | Acc: 0.6296 | F1: 0.4444 | AUC: 0.7333

Epoch 9 [Train]: 100%|████████████████████| 16/16 [00:06<00:00, 2.42it/s]

Epoch 9 [Val]: 100%|████████████████████| 4/4 [00:01<00:00, 2.25it/s]

Epoch 9 Summary:

Train Loss: 0.3844 | Acc: 0.8203 | F1: 0.7850 | AUC: 0.9073

Val Loss: 0.7045 | Acc: 0.7037 | F1: 0.6923 | AUC: 0.7389

Epoch 10 [Train]: 100%|████████████████████| 16/16 [00:06<00:00, 2.38it/s]

Epoch 10 [Val]: 100%|████████████████████| 4/4 [00:01<00:00, 2.25it/s]

Epoch 10 Summary:

Train Loss: 0.3082 | Acc: 0.8750 | F1: 0.8431 | AUC: 0.9514

Val Loss: 0.8548 | Acc: 0.4815 | F1: 0.5882 | AUC: 0.7222

Epoch 11 [Train]: 100%|████████████████████| 16/16 [00:06<00:00, 2.45it/s]

Epoch 11 [Val]: 100%|████████████████████| 4/4 [00:01<00:00, 2.34it/s]

Epoch 11 Summary:

Train Loss: 0.3458 | Acc: 0.8750 | F1: 0.8621 | AUC: 0.9325

Val Loss: 0.7243 | Acc: 0.6296 | F1: 0.4444 | AUC: 0.7222

Epoch 12 [Train]: 100%|████████████████████| 16/16 [00:06<00:00,
2.41it/s]
Epoch 12 [Val]: 100%|████████████████████| 4/4 [00:01<00:00,
2.36it/s]

Epoch 12 Summary:

Train Loss: 0.2566 | Acc: 0.8828 | F1: 0.8598 | AUC: 0.9651

Val Loss: 0.7608 | Acc: 0.5926 | F1: 0.3529 | AUC: 0.7333

Epoch 13 [Train]: 100%|████████████████████| 16/16 [00:06<00:00,
2.46it/s]

Epoch 13 [Val]: 100%|████████████████████| 4/4 [00:01<00:00,
2.24it/s]

Epoch 13 Summary:

Train Loss: 0.1625 | Acc: 0.9531 | F1: 0.9444 | AUC: 0.9910

Val Loss: 0.6654 | Acc: 0.7407 | F1: 0.7200 | AUC: 0.7333

Epoch 14 [Train]: 100%|████████████████████| 16/16 [00:06<00:00,
2.38it/s]

Epoch 14 [Val]: 100%|████████████████████| 4/4 [00:01<00:00,
2.24it/s]

Epoch 14 Summary:

Train Loss: 0.1814 | Acc: 0.9375 | F1: 0.9259 | AUC: 0.9938

Val Loss: 0.7633 | Acc: 0.5926 | F1: 0.3529 | AUC: 0.7500

Epoch 15 [Train]: 100%|████████████████████| 16/16 [00:06<00:00,
2.46it/s]

Epoch 15 [Val]: 100%|████████████████████| 4/4 [00:01<00:00,
2.35it/s]

Epoch 15 Summary:

Train Loss: 0.1435 | Acc: 0.9609 | F1: 0.9541 | AUC: 0.9975

Val Loss: 0.6760 | Acc: 0.6296 | F1: 0.5455 | AUC: 0.7500

Training log saved to: ./ABIDEI/training_log.csv

ROC curve saved to: ./ABIDEI/val_roc_curve.png

Training complete.

F1-score: 0.4827586206896552

ROC_AUC-score: 0.4114583333333333

Finished inference on 28 subjects

Accuracy: 46.43%

Tain/Val/Test Split on all sites

--lr 0.0001 --wd 0.0001 --batch 8 --epochs 15 (no layers frozen)

Epoch 1 [Train]: 100%|████████████████████| 87/87 [00:29<00:00, 2.98it/s]

Epoch 1 [Val]: 100%|████████████████████| 19/19 [00:05<00:00, 3.56it/s]

Epoch 1 Summary:

Train Loss: 0.9863 | Acc: 0.5501 | F1: 0.5621 | AUC: 0.5716

Val Loss: 0.9347 | Acc: 0.5135 | F1: 0.4375 | AUC: 0.5163

>>> Saved new best model at epoch 1 with ACC 0.5135

Epoch 2 [Train]: 100%|████████████████████| 87/87 [00:28<00:00, 3.11it/s]

Epoch 2 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 3.84it/s]

Epoch 2 Summary:

Train Loss: 0.8169 | Acc: 0.5588 | F1: 0.5543 | AUC: 0.5795

Val Loss: 0.9472 | Acc: 0.5068 | F1: 0.6054 | AUC: 0.5503

Epoch 3 [Train]: 100%|████████████████████| 87/87 [00:27<00:00, 3.12it/s]

Epoch 3 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 3.87it/s]

Epoch 3 Summary:

Train Loss: 0.7293 | Acc: 0.6096 | F1: 0.6310 | AUC: 0.6322

Val Loss: 0.7653 | Acc: 0.5270 | F1: 0.5270 | AUC: 0.5662

>>> Saved new best model at epoch 3 with ACC 0.5270

Epoch 4 [Train]: 100%|████████████████████| 87/87 [00:27<00:00, 3.12it/s]

Epoch 4 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 3.82it/s]

Epoch 4 Summary:

Train Loss: 0.6905 | Acc: 0.6096 | F1: 0.6141 | AUC: 0.6496

Val Loss: 0.7526 | Acc: 0.6081 | F1: 0.2927 | AUC: 0.5773

>>> Saved new best model at epoch 4 with ACC 0.6081

Epoch 5 [Train]: 100%|████████████████████| 87/87 [00:27<00:00, 3.11it/s]

Epoch 5 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 3.88it/s]

Epoch 5 Summary:

Train Loss: 0.6648 | Acc: 0.6473 | F1: 0.6453 | AUC: 0.6710

```
Val   Loss: 0.7186 | Acc: 0.6014 | F1: 0.3918 | AUC: 0.5700  
Epoch 6 [Train]: 100%|██████████| 87/87 [00:28<00:00,  
3.09it/s]  
Epoch 6 [Val]: 100%|██████████| 19/19 [00:05<00:00,  
3.70it/s]
```

Epoch 6 Summary:

Train Loss: 0.6130 | Acc: 0.6488 | F1: 0.6452 | AUC: 0.7248

Val Loss: 0.7293 | Acc: 0.5473 | F1: 0.5315 | AUC: 0.5805

```
Epoch 7 [Train]: 100%|██████████| 87/87 [00:28<00:00,  
3.09it/s]
```

```
Epoch 7 [Val]: 100%|██████████| 19/19 [00:04<00:00,  
3.93it/s]
```

Epoch 7 Summary:

Train Loss: 0.6178 | Acc: 0.6444 | F1: 0.6434 | AUC: 0.7173

Val	Loss: 0.7226	Acc: 0.5203	F1: 0.4228	AUC: 0.5585
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```
Epoch 8 [Train]: 100%|██████████| 87/87 [00:27<00:00,  
3.11it/s]
```

```
Epoch 8 [Val]: 100%|██████████████████████| 19/19 [00:04<00:00,  
3.98it/s]
```

Epoch 8 Summary:

Train Loss: 0.6012 | Acc: 0.6807 | F1: 0.6866 | AUC: 0.7404

Val	Loss: 0.7241	Acc: 0.6081	F1: 0.3256	AUC: 0.5801
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```
Epoch 9 [Train]: 100%|███████████| 87/87 [00:27<00:00,  
3.12it/s]
```

```
Epoch 9 [Val]: 100%|███████████| 19/19 [00:04<00:00,  
3.90it/s]
```

Epoch 9 Summary:

Train Loss: 0.5392 | Acc: 0.7431 | F1: 0.7475 | AUC: 0.8078

Val	Loss: 1.0443	Acc: 0.4595	F1: 0.5960	AUC: 0.5949
-----	--------------	-------------	------------	-------------

```
Epoch 10 [Train]: 100%|██████████| 87/87 [00:27<00:00,  
3.12it/s]
```

```
Epoch 10 [Val]: 100%|██████████| 19/19 [00:04<00:00,  
4.07it/s]
```

Epoch 10 Summary:

Train Loss: 0.5378 | Acc: 0.7344 | F1: 0.7382 | AUC: 0.8064

Val	Loss: 0.7859	Acc: 0.5135	F1: 0.5500	AUC: 0.5778
-----	--------------	-------------	------------	-------------

```
Epoch 11 [Train]: 100%|██████████| 87/87 [00:28<00:00,  
3.11it/s]
```

```
Epoch 11 [Val]: 100%|██████████| 19/19 [00:04<00:00,  
3.91it/s]
```

Epoch 11 Summary:

Train Loss: 0.4900 | Acc: 0.7721 | F1: 0.7721 | AUC: 0.8466

Val	Loss: 0.7520	Acc: 0.5811	F1: 0.4151	AUC: 0.5775
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Epoch 12 [Train]: 100%|████████████████████| 87/87 [00:28<00:00, 3.10it/s]
Epoch 12 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 3.88it/s]

Epoch 12 Summary:

Train Loss: 0.4558 | Acc: 0.7736 | F1: 0.7739 | AUC: 0.8714

Val Loss: 0.7999 | Acc: 0.5541 | F1: 0.4000 | AUC: 0.5668

Epoch 13 [Train]: 100%|████████████████████| 87/87 [00:27<00:00, 3.11it/s]

Epoch 13 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 3.89it/s]

Epoch 13 Summary:

Train Loss: 0.4437 | Acc: 0.7983 | F1: 0.8023 | AUC: 0.8786

Val Loss: 1.2367 | Acc: 0.5811 | F1: 0.0312 | AUC: 0.5308

Epoch 14 [Train]: 100%|████████████████████| 87/87 [00:27<00:00, 3.12it/s]

Epoch 14 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 3.91it/s]

Epoch 14 Summary:

Train Loss: 0.4365 | Acc: 0.8142 | F1: 0.8123 | AUC: 0.8817

Val Loss: 1.0384 | Acc: 0.6014 | F1: 0.1449 | AUC: 0.5381

Epoch 15 [Train]: 100%|████████████████████| 87/87 [00:28<00:00, 3.10it/s]

Epoch 15 [Val]: 100%|████████████████████| 19/19 [00:04<00:00, 3.93it/s]

Epoch 15 Summary:

Train Loss: 0.3763 | Acc: 0.8273 | F1: 0.8263 | AUC: 0.9189

Val Loss: 0.9533 | Acc: 0.6149 | F1: 0.2192 | AUC: 0.5503

>>> Saved new best model at epoch 15 with ACC 0.6149

Training log saved to: ./ABIDEI/training_log.csv

ROC curve saved to: ./ABIDEI/val_roc_curve.png

Training complete.

F1-score: 0.21951219512195122

ROC_AUC-score: 0.5937916587316702

Finished inference on 148 subjects

Accuracy: 56.76%

Updated Data Train/Val/Test Split on all sites on MedNet3D Resnet18_23dataset

--lr 0.0001--batch 8 --epochs 15 (layer 4 frozen)

epoch 1
epoch 1 average loss: 0.7894
Epoch time duration: 50.0543212890625
current epoch: 1 current accuracy: 0.4189 best accuracy: 0.4189 at
epoch 1

epoch 2
epoch 2 average loss: 0.7103
Epoch time duration: 45.82671666145325
current epoch: 2 current accuracy: 0.5811 best accuracy: 0.5811 at
epoch 2

epoch 3
epoch 3 average loss: 0.7059
Epoch time duration: 46.38071322441101
current epoch: 3 current accuracy: 0.4595 best accuracy: 0.5811 at
epoch 2

epoch 4
epoch 4 average loss: 0.7029
Epoch time duration: 45.95577359199524
current epoch: 4 current accuracy: 0.5338 best accuracy: 0.5811 at
epoch 2

epoch 5
epoch 5 average loss: 0.7028
Epoch time duration: 46.067646503448486
current epoch: 5 current accuracy: 0.5270 best accuracy: 0.5811 at
epoch 2

epoch 6
epoch 6 average loss: 0.7019
Epoch time duration: 45.53186893463135
current epoch: 6 current accuracy: 0.5270 best accuracy: 0.5811 at
epoch 2

epoch 7
epoch 7 average loss: 0.6993

Epoch time duration: 46.11419486999512
current epoch: 7 current accuracy: 0.5743 best accuracy: 0.5811 at epoch 2

epoch 8
epoch 8 average loss: 0.7078
Epoch time duration: 45.71909523010254
current epoch: 8 current accuracy: 0.5338 best accuracy: 0.5811 at epoch 2

epoch 9
epoch 9 average loss: 0.6992
Epoch time duration: 45.97823786735535
current epoch: 9 current accuracy: 0.5405 best accuracy: 0.5811 at epoch 2

epoch 10
epoch 10 average loss: 0.7093
Epoch time duration: 46.162644147872925
current epoch: 10 current accuracy: 0.5743 best accuracy: 0.5811 at epoch 2

epoch 11
epoch 11 average loss: 0.6940
Epoch time duration: 45.632129430770874
current epoch: 11 current accuracy: 0.4662 best accuracy: 0.5811 at epoch 2

epoch 12
epoch 12 average loss: 0.6957
Epoch time duration: 46.145867586135864
current epoch: 12 current accuracy: 0.5068 best accuracy: 0.5811 at epoch 2

epoch 13
epoch 13 average loss: 0.6964
Epoch time duration: 45.80393195152283
current epoch: 13 current accuracy: 0.5203 best accuracy: 0.5811 at epoch 2

epoch 14
epoch 14 average loss: 0.6939
Epoch time duration: 45.85673451423645
current epoch: 14 current accuracy: 0.4865 best accuracy: 0.5811 at epoch 2

epoch 15
epoch 15 average loss: 0.6992
Epoch time duration: 46.06570744514465

current epoch: 15 current accuracy: 0.5743 best accuracy: 0.5811 at
epoch 2
train completed, best_metric: 0.5811 at epoch: 2

**Tain/Val Split on all sites except NYU (test) --lr 0.0001
--wd 0.0001 --batch 8 --epochs 15 (no layers frozen)**

Epoch 14 Summary:

Train Loss: 0.4254 | Acc: 0.7973 | F1: 0.7925 | AUC: 0.8887

Val Loss: 0.7792 | Acc: 0.5959 | F1: 0.6144 | AUC: 0.6151

>>> Saved new best model at epoch 14 with AUC 0.6151

TEST SET:

183 subjects

F1-score: 0.35384615384615387

ROC_AUC-score: 0.5242210321324245

Accuracy: 54.10%

**Tain/Val/Test Split on only NYU --lr 0.0001 --wd 0.0001
--batch 8 --epochs 15 (no layers frozen)**

Epoch 6 Summary:

Train Loss: 0.5145 | Acc: 0.7266 | F1: 0.6465 | AUC: 0.8204

Val Loss: 0.6506 | Acc: 0.7407 | F1: 0.7200 | AUC: 0.7111

>>> Saved new best model at epoch 6 with ACC 0.7407

TEST SET:

28 subjects

F1-score: 0.4827586206896552

ROC_AUC-score: 0.4114583333333333

Accuracy: 46.43%

**Tain/Val/Test Split on all sites --lr 0.0001 --wd 0.0001 --
batch 8 --epochs 15 (no layers frozen)**

Epoch 4 Summary:

Train Loss: 0.6905 | Acc: 0.6096 | F1: 0.6141 | AUC: 0.6496

Val Loss: 0.7526 | Acc: 0.6081 | F1: 0.2927 | AUC: 0.5773

>>> Saved new best model at epoch 4 with ACC 0.6081

TEST SET:

148 subjects

F1-score: 0.21951219512195122

ROC_AUC-score: 0.5937916587316702

Accuracy: 56.76%

Updated Data Tain/Val/Test Split on all sites on MedNet3D Resnet18_23dataset --lr 0.0001--batch 8 -- epochs 15 (layer 4 frozen)

epoch 2 average loss: 0.7103

Epoch time duration: 45.82671666145325

current epoch: 2 current accuracy: 0.5811 **best accuracy: 0.5811 at epoch 2**

***using resnet10 had a lower accuracy overall and it's accuracy only went down after the first epoch

Tain/Val Split on all sites except NYU (test) --lr 0.0001 --wd 0.0001 --batch 8 --epochs 15 (no layers frozen)

Epoch 14 Summary:

Train Loss: 0.4254 | Acc: 0.7973 | F1: 0.7925 | AUC: 0.8887

Val Loss: 0.7792 | Acc: 0.5959 | F1: 0.6144 | AUC: 0.6151

>>> Saved new best model at epoch 14 with AUC 0.6151

TEST SET:

183 subjects

F1-score: 0.35384615384615387

ROC_AUC-score: 0.5242210321324245

Accuracy: 54.10%